

Name: _____



Priority of operations with positive and negative integers, decimals and fractions

Task A: Reviewing priority of operations

1) Work out the following using the priority of operations, giving your answer in its simplest form where possible.

a) $\frac{9}{10} + \frac{3}{4} \times \frac{2}{5}$

b) $\frac{11}{15} - \frac{3}{5} \div \frac{9}{10}$

c) $\frac{9}{10} \times \frac{1}{2} - \frac{3}{5} \div \frac{4}{5}$

2) Work out the following using the priority of operations, giving your answer in its simplest form and a mixed number where possible.

a) $2\frac{1}{2} + 3\frac{2}{3} \times \frac{3}{4}$

b) $2\frac{1}{4} \times \frac{4}{5} - \frac{3}{5} \div 1\frac{3}{10}$



Task B: Powers and roots of fractions

1) Fill in the blanks to complete the calculation

$$\frac{3}{5} - \frac{4}{5} \times \frac{2}{3} + \left(\frac{3}{5}\right)^2$$

$$\frac{3}{5} - \frac{4}{5} \times \frac{2}{3} + \boxed{}$$

$$\frac{3}{5} - \boxed{} + \boxed{}$$

$$\frac{3}{5} + \boxed{} + \left(-\boxed{}\right)$$

$$\frac{\boxed{}}{150} + \frac{\boxed{}}{150} + \left(-\frac{\boxed{}}{150}\right) = \boxed{} = \boxed{}$$

2) Work out the answer to the following, giving your answer in its simplest form.

$$\frac{9}{10} \times \frac{4}{5} - \frac{2}{3} \times \sqrt[3]{\frac{8}{27}}$$

3) Lucas has made an error.

Find the error and correctly complete the calculation

$$\frac{2}{3} + \sqrt{6\frac{1}{4}} \times \frac{3}{5}$$

$$\frac{2}{3} + \sqrt{6} + \frac{1}{2} \times \frac{3}{5}$$

$$\frac{2}{3} + \sqrt{6} + \frac{3}{10}$$

$$\sqrt{6} + \frac{2}{3} + \frac{3}{10} =$$

$$\sqrt{6} + \frac{20}{30} + \frac{9}{30} = \sqrt{6} \frac{29}{30}$$

Task C: Powers and roots of fractions

1) Fill in the blanks to complete the calculation

$$2 - \left(\frac{2}{5}\right)^2 \times \sqrt{\frac{49}{100}}$$

$$2 - \boxed{} \times \boxed{}$$

$$2 - \frac{\boxed{}}{250}$$

$$\frac{\boxed{}}{250} - \frac{\boxed{}}{250} = \frac{\boxed{}}{250} = \frac{\boxed{}}{125} = 1 \frac{\boxed{}}{125}$$

2) Work out the answer to the following, ensuring to give your answer as a mixed number in its simplest form

$$\text{a) } \frac{9}{10} + (2.4 \div 0.8) \times \frac{2}{3}$$

$$\text{b) } 15 + (3 \div 5) \times (0.3 \div 4)$$

3) Work out the answer to the following, ensuring to give your answer as a mixed number in its simplest form

$$\text{a) } 2 - \frac{3}{4} \times \sqrt{\frac{49}{81}}$$

$$\text{b) } \left(\frac{9}{10}\right)^2 - (2.5 \div 0.5) \times 2$$

